

CYBERPRO AI INTERNSHIP

CyberPro AI Assessment Report

CPAI-2026-007

Bina AI Digital Forensics Investigation

Acquisition Method and Integrity Verification Plan

Prepared By: Britton Smith

Course: CPIN-280 Cyber Security Internship

Platform: CyberProAI Bina AI

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Assessment Status: COMPLETE

Bottom Line Up Front (BLUF)

The CyberProAI Bina AI investigation successfully demonstrated the use of artificial intelligence as an educational and investigative assistant within a digital forensics scenario. A medium-difficulty AI-guided exercise, *Acquisition Method and Integrity Verification Plan*, was selected from the Digital Forensics Investigation learning path. The exercise required development of a professional Digital Evidence Acquisition Plan for a suspected corporate data breach involving deleted evidence. Through AI-guided instruction and independent technical reasoning, appropriate forensic acquisition procedures, evidence preservation practices, and cryptographic integrity verification methods were applied. The submitted response received a **Perfect** evaluation from Bina AI, confirming both technical accuracy and adherence to recognized digital forensic best practices.

1. Assessment Background

As part of the CyberProAI internship curriculum, students were introduced to the Bina AI cybersecurity learning platform to evaluate how artificial intelligence can support technical education and investigative decision-making. Unlike the Breaking Barriers penetration testing assessments, which focused on offensive security operations, this exercise emphasized digital forensic methodology and evidence preservation following a simulated security incident.

The selected exercise, **Acquisition Method and Integrity Verification Plan**, presented a realistic corporate investigation involving a powered-down laptop believed to contain a deleted malicious script associated with a data breach. Rather than immediately requesting a solution, Bina AI first introduced the forensic concepts necessary to complete the investigation before requiring practical application. This assessment documents the investigative process, technical findings, AI-assisted learning experience, and lessons gained throughout the exercise.

2. Assessment Objectives

The Digital Forensics Investigation was performed to accomplish the following professional objectives:

AI-Assisted Technical Learning: Evaluate the effectiveness of artificial intelligence as a cybersecurity educational resource while maintaining independent technical reasoning.

Evidence Acquisition Planning: Determine the most appropriate forensic acquisition methodology based upon the investigative scenario.

Integrity Verification: Apply accepted cryptographic validation techniques to preserve evidentiary integrity throughout the acquisition process.

Professional Documentation: Produce a structured Digital Evidence Acquisition Plan reflecting recognized digital forensic best practices.

Continuous documentation and screenshot collection were maintained throughout the investigation to preserve a complete educational and technical record.

3. Technical Findings & Validation

Finding 1 - Physical Evidence Acquisition Selected

Analysis

The investigation involved a corporate laptop suspected of containing a deleted malicious script. Because deleted files frequently remain recoverable within unallocated storage space until overwritten, selecting an acquisition method capable of preserving every sector of the storage device was essential.

Validation

A **Physical Acquisition** was selected as the appropriate forensic methodology because it captures a complete bit-for-bit image of the storage media, including deleted files, file system metadata, slack space, and unallocated sectors that may contain evidentiary artifacts unavailable through logical acquisition methods.

Finding 2 - Cryptographic Integrity Verification

Analysis

Digital evidence possesses little value if investigators cannot demonstrate that it remained unchanged throughout acquisition and analysis. Maintaining forensic integrity is therefore a foundational requirement of any digital forensic investigation.

Validation

The submitted acquisition plan recommended generating **SHA-256 cryptographic hash values** for both the original storage device and the forensic image immediately following acquisition. Matching hash values mathematically verify that the forensic image is an exact duplicate of the original evidence while preserving legal admissibility and investigative integrity.

Finding 3 - AI-Assisted Forensic Education

Analysis

Rather than simply generating a solution, Bina AI presented instructional material explaining physical versus logical acquisition, forensic integrity principles, and cryptographic verification before requesting an investigative response.

Validation

The completed Digital Evidence Acquisition Plan successfully integrated these concepts with independent technical reasoning. Upon submission, Bina AI awarded the investigation a **Perfect** evaluation, confirming that the selected methodology accurately reflected accepted digital forensic practices and professional investigative standards.

4. Professional Remediation Framework

Level 1: Immediate Recommendations (Technical Development)

Continue completing AI-guided digital forensic exercises to reinforce evidence acquisition methodology, cryptographic validation procedures, and investigative documentation standards.

Expand practical familiarity with forensic terminology including chain of custody, write-blocking, forensic imaging, and evidence preservation procedures.

Level 2: Tactical Recommendations (Operational Skill Development)

Transition from AI-guided exercises into practical laboratory environments utilizing professional forensic software capable of performing disk imaging, evidence recovery, and file system analysis.

Develop familiarity with common forensic workflows used by Security Operations Centers (SOC), Incident Response teams, and Digital Forensics and Incident Response (DFIR) professionals.

Level 3: Strategic Recommendations (Long-Term Professional Growth)

Incorporate artificial intelligence into future cybersecurity education as a supplemental analytical assistant while continuing to independently validate technical recommendations against recognized industry standards.

Expand future studies into memory forensics, live response investigations, malware analysis, and enterprise incident response to build a comprehensive understanding of digital forensic operations.

5. Challenges Encountered & AI Learning Process

Challenge 1 - Understanding Evidence Acquisition Methodology

Challenge

Prior to this exercise, formal digital forensic acquisition procedures represented unfamiliar technical subject matter.

Resolution

Bina AI introduced the concepts of physical and logical acquisition through structured instructional content before requiring practical application, allowing technical understanding to develop progressively throughout the investigation.

Challenge 2 - Applying Cryptographic Integrity Principles

Challenge

While cryptographic hashing was already familiar as a cybersecurity concept, applying it specifically to forensic evidence preservation required additional understanding.

Resolution

Instruction explaining SHA-256 integrity verification demonstrated how cryptographic hashes provide mathematical proof that forensic evidence remains unchanged from acquisition through courtroom presentation, reinforcing the legal significance of evidence preservation.

Challenge 3 - Validating AI Recommendations

Challenge

Artificial intelligence should not replace professional judgment during cybersecurity investigations.

Resolution

Rather than accepting AI-generated guidance without evaluation, recommendations were reviewed, understood, and incorporated only after confirming the technical reasoning behind each forensic decision. This reinforced the importance of treating AI as an investigative assistant rather than an authoritative source.

6. Lessons Learned & Conclusion

The successful completion of this Digital Forensics Investigation demonstrates that artificial intelligence can significantly accelerate cybersecurity education when paired with independent technical analysis and structured investigative thinking.

The exercise strengthened understanding of evidence acquisition planning, forensic integrity, cryptographic validation, and documentation standards while illustrating how professional investigators preserve digital evidence throughout the investigative lifecycle. Equally important, the investigation reinforced that AI serves most effectively as an educational resource capable of explaining technical concepts, organizing information, and encouraging analytical reasoning rather than replacing professional expertise.

Completion of the exercise with a **Perfect** evaluation confirms successful application of accepted digital forensic principles and demonstrates measurable growth in cybersecurity knowledge beyond offensive security operations. This assessment broadens the internship portfolio by documenting practical exposure to digital forensics, an essential discipline supporting incident response, legal investigations, and enterprise security operations.

Appendix A – Exercise Solution

DIGITAL EVIDENCE ACQUISITION PLAN

1. ACQUISITION METHOD SELECTION

Chosen Method: Physical Acquisition

Technical Justification:

A physical acquisition is required because investigators suspect the threat actor executed a custom script and then deleted it to conceal evidence. Deleted files frequently remain within unallocated space after the operating system removes the file system reference, making them recoverable through a complete bit-stream acquisition. A logical acquisition would collect only active files recognized by the operating system and could permanently exclude deleted artifacts that may be critical to the investigation.

Prior to acquisition, the evidence should be photographed and documented, identifying information such as serial numbers and asset tags should be recorded, and chain-of-custody procedures should be established. Because the laptop is already powered down, the storage device should remain offline and be acquired using a dedicated forensic workstation or write-blocking device to prevent accidental modification or execution of malicious software.

2. FORENSIC INTEGRITY & VERIFICATION

Integrity Pillar: Integrity

Technical Action:

Following acquisition, cryptographic hash values (such as SHA-256) should be generated for both the original storage media and the forensic image. Matching hash values mathematically verify that the forensic image is an exact, unaltered duplicate of the original evidence. This verification preserves forensic integrity and provides defensible proof that the evidence presented during the investigation is the same evidence collected at the scene.

Appendix B – Figure Index

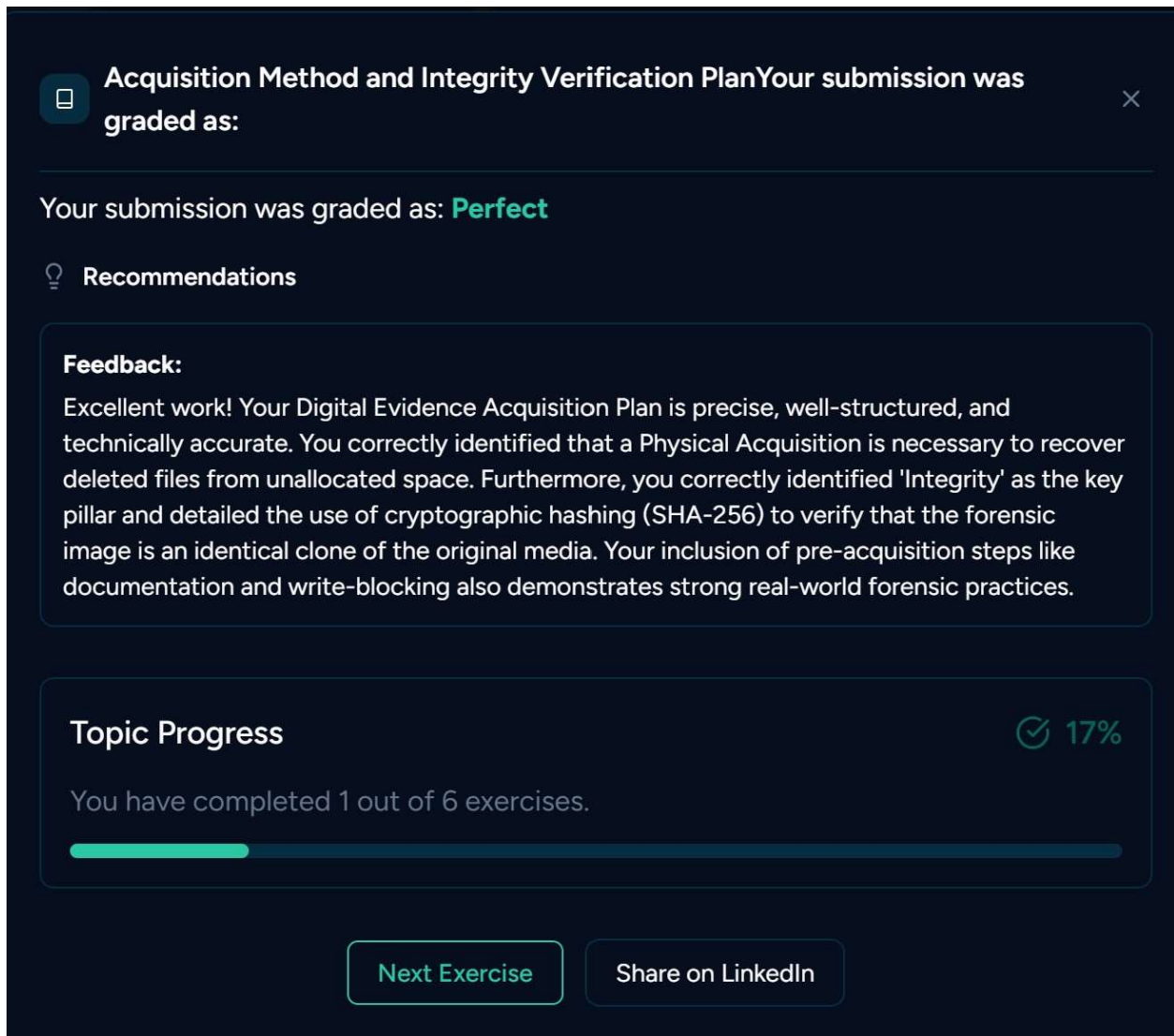


Figure 1. Exercise Evaluation and Assessment Results.

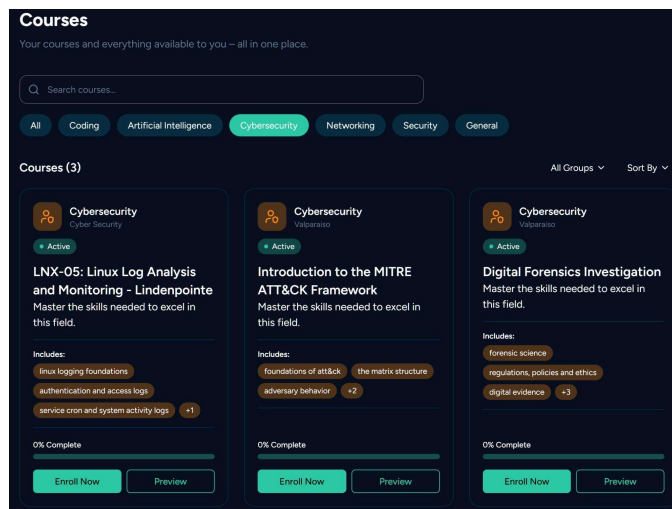


Figure 2. CyberProAI Course Dashboard.

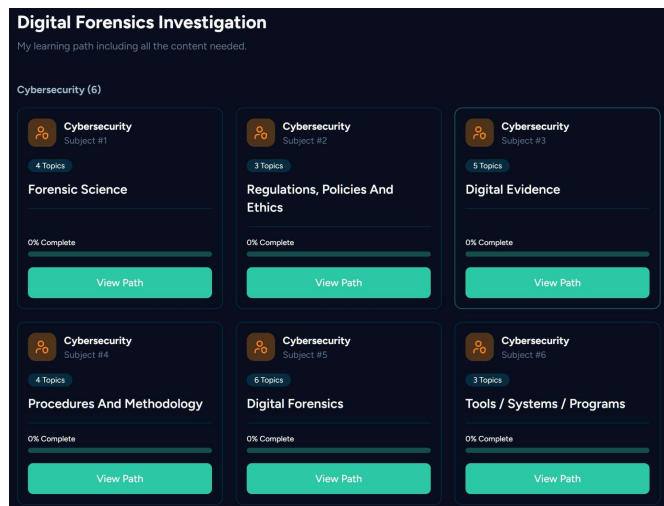


Figure 3. Digital Forensics Investigation Learning Path.

Acquisition Method And Integrity Verification Plan

Forensic-Science Medium

Goal

Scenario:

You have been dispatched to investigate a corporate laptop linked to a suspected data breach. Investigators believe the threat actor executed a custom script and subsequently deleted it from the drive to cover their tracks. The laptop is currently powered down and secured in your lab.

As the lead investigator, you must draft a brief **Digital Evidence Acquisition Plan** in the editor before proceeding. Your plan must address the following two points based on forensic acquisition best practices:

State whether you should perform a **Physical Acquisition** or a **Logical Acquisition**, and explain why this choice is necessary for this specific case.

Identify which of the **Three Pillars of Forensic Integrity** ensures the image is an identical clone, and describe the specific technical action you must take to satisfy this rule.

Provide a brief, clear response for both points in the text area below.

[💡 Learn Concept](#)

Your Response

Write your answer here...

Figure 4. Acquisition Method and Integrity Verification Plan Exercise.

Understanding Forensic Acquisition & Integrity

When responding to a security incident involving deleted evidence, the methods you use to collect data can make or break your investigation. To draft an effective Digital Evidence Acquisition Plan, you must understand how data behaves on storage media and how to mathematically prove that your evidence remains unaltered.

1. Physical vs. Logical Acquisition

To choose the correct acquisition method, you must distinguish between what the operating system sees and what the physical drive actually contains:

Logical Acquisition: Captures only the active, accessible files and directories currently recognized by the file system. It ignores deleted space, file slack, and unallocated clusters.

Physical Acquisition: Performs a bit-stream, sector-by-sector copy of the entire physical storage medium. This captures everything—including active files, system metadata, slack space, and **unallocated space** where deleted files often reside before they are overwritten.

When a threat actor deletes a file, the operating system typically only removes its pointer, leaving the actual file contents intact in unallocated space. Selecting the wrong acquisition type can result in permanent loss of this critical artifact.

💡 **Analogy:** Think of a physical acquisition like taking a plaster cast of an entire footprint, capturing every microscopic crack and soil grain. A logical acquisition is like drawing a sketch of just the shoe's logo; it captures the obvious design but misses the hidden details embedded in the ground.

2. Proving Forensic Integrity

In digital forensics, you must prove that the evidence analyzed in the lab is an exact, unaltered duplicate of the source media. This is governed by the pillar of **Integrity**.

To satisfy this requirement, investigators rely on mathematical validation. Before and during the imaging process, a forensic tool runs a mathematical algorithm over the source drive to generate a unique digital fingerprint. If even a single bit of data changes on the target image, the fingerprints will not match.

Real-world tie: In corporate espionage cases, examiners routinely rely on physical acquisitions to recover wiped registry keys or deleted malware payloads from unallocated clusters, using cryptographic verification to ensure the evidence stands up in court.

Figure 5. AI Instructional Guidance.